

ECON 4720: HA03

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Problems

Problem 1	2
1a	2
1b	2
1c	3
1d	4
Problem 2	5
2a	5
2b	6
2c	6
2d	7
2e	8
2f	10
2g	10
2h	11
Problem 3	11
3a	11
3b	14
3c	16
3d	18
Problem 4	18
4a	18
4a	18

Problem 1

Suppose that $Y_1, Y_2, \dots, Y_n$ are random variables with a common mean μ_Y , a common variance σ_Y^2 , and the same correlation ρ (so that the correlation between Y_i and Y_j is equal to ρ for all pairs $i \neq j$).

1a

Show that $\text{Cov}(Y_i, Y_j) = \rho\sigma_Y^2$ for $i \neq j$.

Solution Attempt: We know that correlation is defined as

$$\text{Corr}(Y_i, Y_j) = \frac{\text{Cov}(Y_i, Y_j)}{\sqrt{\text{Var}(Y_i)\text{Var}(Y_j)}}$$

and from our preamble we also know that “the correlation between Y_i and Y_j is equal to ρ for all pairs $i \neq j$ ” and common variance implying $\text{Var}(Y_i) = \text{Var}(Y_j) = \sigma_Y^2$. Thus,

$$\begin{aligned}\rho &= \frac{\text{Cov}(Y_i, Y_j)}{\sqrt{\sigma_Y^2 \sigma_Y^2}} \\ &= \frac{\text{Cov}(Y_i, Y_j)}{\sigma_Y^2} \\ \rho\sigma_Y^2 &= \text{Cov}(Y_i, Y_j)\end{aligned}$$

which shows what the question asks.

□

1b

Suppose that $n = 2$. Show that $E(\bar{Y}) = \mu_Y$ and $\text{Var}(\bar{Y}) = \frac{1}{2}\sigma_Y^2 + \frac{1}{2}\rho\sigma_Y^2$.

Solution Attempt: This change brings our random variable set to $\{Y_1, Y_2\}$. First, we find $E(\bar{Y})$. Note that $\bar{Y}$ is

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i = \frac{1}{2}(Y_1 + Y_2)$$

Thus,

$$E(\bar{Y}) = E\left(\frac{1}{2}(Y_1 + Y_2)\right) = \frac{1}{2}(E(Y_1) + E(Y_2))$$

From the preamble, we know that $E(Y_i) = E(Y_j) = \mu_Y$, thus,

$$E(\bar{Y}) = \frac{1}{2}(2\mu_Y) = \mu_Y$$

Now, for the variance, $Var(\bar{Y})$. From earlier, we know

$$\begin{aligned} Var(\bar{Y}) &= Var\left(\frac{1}{2}(Y_1 + Y_2)\right) = \frac{1}{4}Var(Y_1 + Y_2) \\ &= \frac{1}{4}(Var(Y_1) + Var(Y_2) + 2Cov(Y_1, Y_2)) && \text{Expansion} \\ &= \frac{1}{4}(2\sigma_Y^2 + 2\rho\sigma_Y^2) && \text{From preamble and (a)} \\ &= \frac{1}{2}\mu_Y^2 + \frac{1}{2}\rho\sigma_Y^2 && \text{Shows what we want} \end{aligned}$$

□

1c

For $n \geq 2$, show that $E(\bar{Y}) = \mu_Y$ and $Var(\bar{Y}) = \frac{\sigma_Y^2}{n} + \left(\frac{n-1}{n}\right)\rho\sigma_Y^2$.

Solution Attempt: Now our set of random variables is $\{Y_1, Y_2, \dots, Y_n\}$. The expectation of $\bar{Y}$ follows a similar process as with (b).

$$\begin{aligned} \bar{Y} &= \frac{1}{n}(Y_1 + Y_2 + \dots + Y_n) = \frac{1}{n} \cdot n\mu_Y = \mu_Y \\ E(\bar{Y}) &= E\left(\frac{1}{n}(Y_1 + Y_2 + \dots + Y_n)\right) = \frac{1}{n}(\underbrace{E(Y_1) + E(Y_2) + \dots + E(Y_n)}_{n \text{ terms}}) \\ &= \frac{1}{n}(n \cdot \mu_Y) = \mu_Y \end{aligned}$$

Note that we used the same principle that $E(Y_i) = E(Y_j) = \mu_Y$ to accomplish the above. Now to

show the variance of $\bar{Y}$.

$$\begin{aligned}
Var(\bar{Y}) &= Var\left(\frac{1}{n}(Y_1 + Y_2 + \cdots + Y_n)\right) = \frac{1}{n^2}Var(Y_1 + Y_2 + \cdots + Y_n) \\
&= \frac{1}{n^2}\left(\overbrace{Var(Y_1) + Var(Y_2) + \cdots + Var(Y_n)}^{n \text{ terms}} + \overbrace{2Cov(Y_1, Y_2) + \cdots + 2Cov(Y_1, Y_n)}^{n \text{ choose 2 terms}}\right) \\
&= \frac{1}{n^2} \cdot n\sigma_Y^2 + \frac{1}{n^2} \binom{n}{2} 2\rho\sigma_Y^2 \\
&= \frac{\sigma_Y^2}{n} + \frac{1}{n^2} \cdot \frac{n(n-1)}{2} 2\rho\sigma_Y^2 \\
&= \frac{\sigma_Y^2}{n} + \frac{n-1}{n} \rho\sigma_Y^2
\end{aligned}$$

We have n choose 2 terms because this is every pairwise correlation and that's how many there exist. Otherwise, the rest is straightforward and we showcase what we are looking for.

□

1d

When n is very large, show that $Var(\bar{Y}) \approx \rho\sigma_Y^2$.

Solution Attempt: Take our expression from above,

$$\begin{aligned}
Var(\bar{Y}) &= \frac{\sigma_Y^2}{n} + \left(\frac{n-1}{n}\right) \rho\sigma_Y^2 \\
\frac{\sigma_Y^2}{n} &\rightarrow 0 \quad \text{as } n \rightarrow \infty \\
\frac{n-1}{n} &\rightarrow 1 \quad \text{as } n \rightarrow \infty
\end{aligned}$$

$Var(\bar{Y}) \rightarrow \rho\sigma_Y^2$ for large n and $Var(\bar{Y}) \approx \rho\sigma_Y^2$.

□

Problem 2

On the companion website, you will find the spreadsheet `Age_HourlyEarnings`, which contains the joint distribution of age (Age) and average hourly earnings (AHE) for 25-to-34 year old full-time workers in 2012 with an education level that exceeds a high school diploma. Use this joint distribution to carry out the following exercises.

2a

Compute the marginal distribution of Age.

Solution Attempt: For this part and future parts, I've converted the dataset into a clean `.csv` file and saved it as `df`.

```
df.sum(axis=0)
```

```
25    0.084967
26    0.090789
27    0.085471
28    0.093518
29    0.103331
30    0.104217
31    0.103166
32    0.108084
33    0.108386
34    0.114542
dtype: float64
```

This above shows us the values of the Age marginal distribution. Notice that this sums to one.

```
age_marginal = df.sum(axis=0)
age_marginal.sum()
```

```
0.9964715700000001
```

□

2b

Compute the mean of AHE for each value of Age; that is, compute $E(\text{AHE} \mid \text{Age} = 25)$ and so forth.

Solution Attempt: For each age $a \in \{25, \dots, 34\}$

$$E(\text{AHE} \mid \text{Age} = a) = \sum_y y \cdot P(\text{AHE} = y \mid \text{Age} = a) = \frac{\sum_y y \cdot P(\text{AHE} = y, \text{Age} = a)}{\sum_y P(\text{AHE} = y, \text{Age} = a)}$$

```
means = {}
for age in df.columns:
    col = df[age]
    num = (col.index * col).sum()
    den = col.sum()
    means[age] = num / den

age_means = pd.DataFrame(
    list(means.items()),
    columns = ['Age', 'E(AHE | Age)']
)
print(age_means)
```

	Age	E(AHE Age)
0	25	17.634440
1	26	19.019161
2	27	19.704933
3	28	20.182865
4	29	21.142915
5	30	21.811767
6	31	22.724216
7	32	23.691860
8	33	23.279501
9	34	23.984285

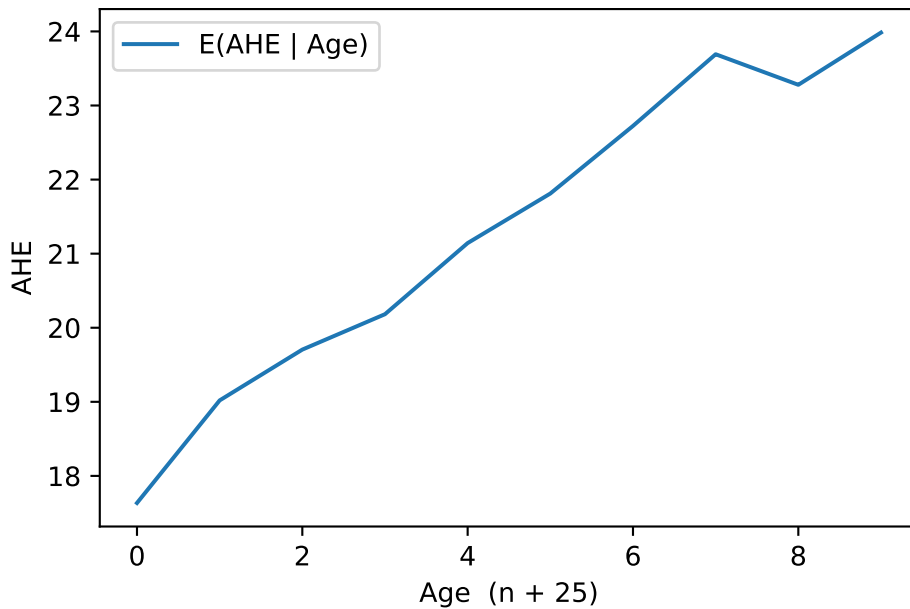
□

2c

Compute and plot the mean of AHE versus Age. Are average hourly earnings and age related? Explain.

Solution Attempt:

```
age_means.plot(
    xlabel = 'Age (n + 25)',
    ylabel = 'AHE'
)
```



□

2d

Use the law of iterated expectations to compute the mean of AHE; that is, compute $E(\text{AHE})$.

Solution Attempt: Here is what we are looking for,

$$E(\text{AHE}) = E(E(\text{AHE} | \text{Age})) = \sum_a \underbrace{E(\text{AHE} | \text{Age} = a)}_{m(a)} \underbrace{P(\text{Age} = a)}_{P_A}$$

where

$$m(a) = \frac{\sum_y y \cdot P(\text{AHE} = y, \text{Age} = a)}{\sum_y P(\text{AHE} = y, \text{Age} = a)} \quad \text{and} \quad P_a = \sum_y P(\text{AHE} = y, \text{Age} = a)$$

And, we can verify directly that

$$E(\text{AHE}) = \sum_a \sum_y y \cdot P(\text{AHE} = y, \text{Age} = a)$$

```
# P(Age=a)
P_age = df.sum(axis=0)

# m(a) = E[AHE | Age=a]
means = {}
for a in df.columns:
    col = df[a]
    means[a] = (col.index * col).sum() / col.sum()
m = pd.Series(means)

# E[AHE] via law of iterated expectations
E_AHE = (m * P_age).sum()
print("E[AHE] via LIE =", E_AHE)
```

E[AHE] via LIE = 21.430079466

```
E_AHE_direct = (df.mul(df.index, axis=0)).sum().sum()
print("E[AHE] direct =", E_AHE_direct)
```

E[AHE] direct = 21.430079466

Great, we get the same things.

□

2e

Compute the variance of AHE.

Solution Attempt: The direct definition,

$$\mu = E(\text{AHE}), \quad E(\text{AHE}^2) = \sum_y y^2 \cdot P(\text{AHE} = y), \quad \boxed{\text{Var}(\text{AHE}) = E(\text{AHE}^2) - \mu^2}$$

Law of total variance says,

$$\boxed{\text{Var}(\text{AHE}) = E(\text{Var}(\text{AHE} \mid \text{Age})) = \text{Var}(E(\text{AHE} \mid \text{Age}))}$$

We can do this directly, or with the law of total variance. Let's try both see if we get the same.


```
# P(AHE = y)
P_ahe = df.sum(axis=1)

# mean and second moment
mu = (P_ahe.index * P_ahe).sum()
E2 = ((P_ahe.index**2) * P_ahe).sum()

var_ahe = E2 - mu**2
print('Var(AHE) =', var_ahe)
```

Var(AHE) = 165.45054164692527

And now with law of total variance,

```
# weights over Age
P_age = df.sum(axis=0)

m, v = {}, {}
for a in df.columns:
    col = df[a]
    p = col / col.sum()
    ma = (p.index * p).sum()
    E2a = ((p.index**2) * p).sum()
    m[a] = ma
    v[a] = E2a - ma**2

m = pd.Series(m)
v = pd.Series(v)

mu = (m * P_age).sum()
var_ahe_LTV = (v * P_age).sum() + ((m - mu)**2 * P_age).sum()
print('Var(AHE) via LTV =', var_ahe_LTV)
```

Var(AHE) via LTV = 163.83011614687115

There seems to be some error in rounding, truncating, or data types but we get essentially the same thing. For grading purposes, go with the first one.

□

2f

Compute the covariance between AHE and Age.

Solution Attempt: Let $Y = \text{AHE}$ and $A = \text{Age}$.

$$\text{Cov}(Y, A) = E(YA) - E(Y)E(A)$$

From the joint table $P(Y = y, A = a)$,

$$E(Y) = \sum_y yP_Y(y), \quad E(A) = \sum_a aP_A(a), \quad E(YA) = \sum_y \sum_a yaP(y, a)$$

```
# I'm going to try to fix the previous data type issue.
# Ensure numeric Age labels
df.columns = df.columns.astype(int)

# marginals
P_ahe = df.sum(1)      # P(Y)
P_age = df.sum(0)      # P(A)

# means
E_Y = (P_ahe.index * P_ahe).sum()
E_A = (P_age.index * P_age).sum()

# cross-moment E(YA) from joint
E_YA = df.mul(df.columns, 1).mul(df.index, 0).to_numpy().sum()

# covariance
cov_YA = E_YA - E_Y * E_A
print('Cov(AHE, Age) =', cov_YA)
```

$\text{Cov}(\text{AHE}, \text{Age}) = 7.855846033653734$

□

2g

Compute the correlation between AHE and Age.

Solution Attempt:

```

Var_Y = ((P_ahe.index ** 2) * P_ahe).sum() - E_Y ** 2
Var_A = ((P_age.index ** 2) * P_age).sum() - E_A ** 2
corr = cov_YA / (Var_Y ** 0.5 * Var_A ** 0.5)
print('Corr(AHE, Age) =', corr)

```

Corr(AHE, Age) = 0.18226305899108558

□

2h

Relate your answers in parts (f) and (g) to the plot you constructed in (c).

Solution Attempt: The plot in (c) shows $E(\text{AHE} \mid \text{Age})$ rising roughly from \$18 at 25 to roughly \$24 at 34, so AHE and Age (and a positive correlation, albeit not huge).

Your variance result and the law of total variance say

$$\text{Var}(\text{AHE}) = E(\text{Var}(\text{AHE} \mid \text{Age})) + \text{Var}(E(\text{AHE} \mid \text{Age})).$$

The plot reflects the second term (the between-age spread of the conditional means), which is small relative to the total variance—so most dispersion in AHE within age groups, not across ages.

So, basically, the plot explains the direction (positive association), while the variance/cov explain the magnitude.

□.

Problem 3

Let $X_1, \dots, X_n$ be a random sample from the Bernoulli distribution with parameter $p = 0.78$ and $n = 25$. Generate $M = 100$ such random draws (there will be $M \times n = 2500$ random variables generated). For each random draw calculate a sample mean $\bar{X}^{(j)}$, where $j = 1, \dots, M$.

3a

Draw a histogram $\bar{X}^{(1)}, \dots, \bar{X}^{(M)}$. Repeat the same for $n = 100$. Discuss the Law of Large Numbers.

Solution Attempt:

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# parameters
p = 0.78
M = 100
rng = np.random.default_rng(42)

def simulate_means(n, M, p, rng):
    draws = rng.binomial(1, p, size = (M, n))
    return draws.mean(1)

mean_25 = simulate_means(25, M, p, rng)
mean_100 = simulate_means(100, M, p, rng)

plt.figure(figsize = (7,4))
plt.hist(mean_25, bins = 10, )
plt.xlabel('Sample mean (n = 25)')
plt.ylabel('Frequency')
plt.title('Hist of Sample Means (Ber: p = 0.78, n = 25, M = 100)')

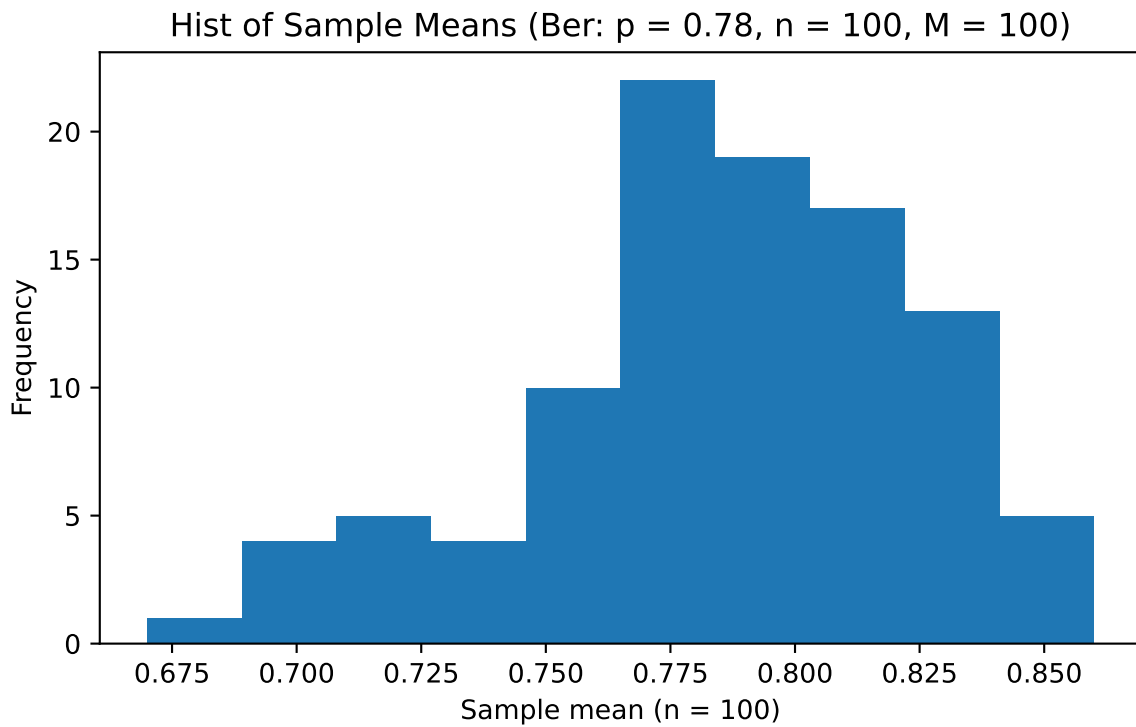
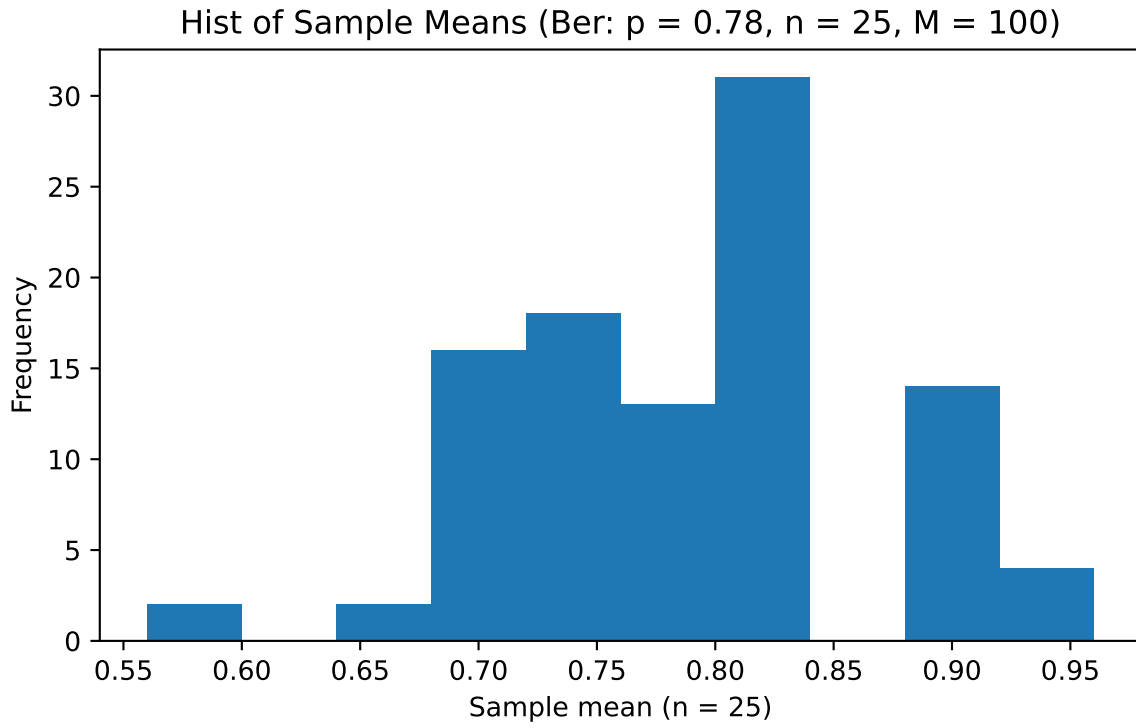
plt.figure(figsize = (7,4))
plt.hist(mean_100, bins = 10)
plt.xlabel('Sample mean (n = 100)')
plt.ylabel('Frequency')
plt.title('Hist of Sample Means (Ber: p = 0.78, n = 100, M = 100)')

```

```

Text(0.5, 1.0, 'Hist of Sample Means (Ber: p = 0.78, n = 100, M = 100)')

```



□

3b

For $n = 25$ and $n = 100$ draw corresponding histograms to illustrate the Central Limit Theorem. Discuss.

Solution Attempt:

```
from math import sqrt, pi, exp

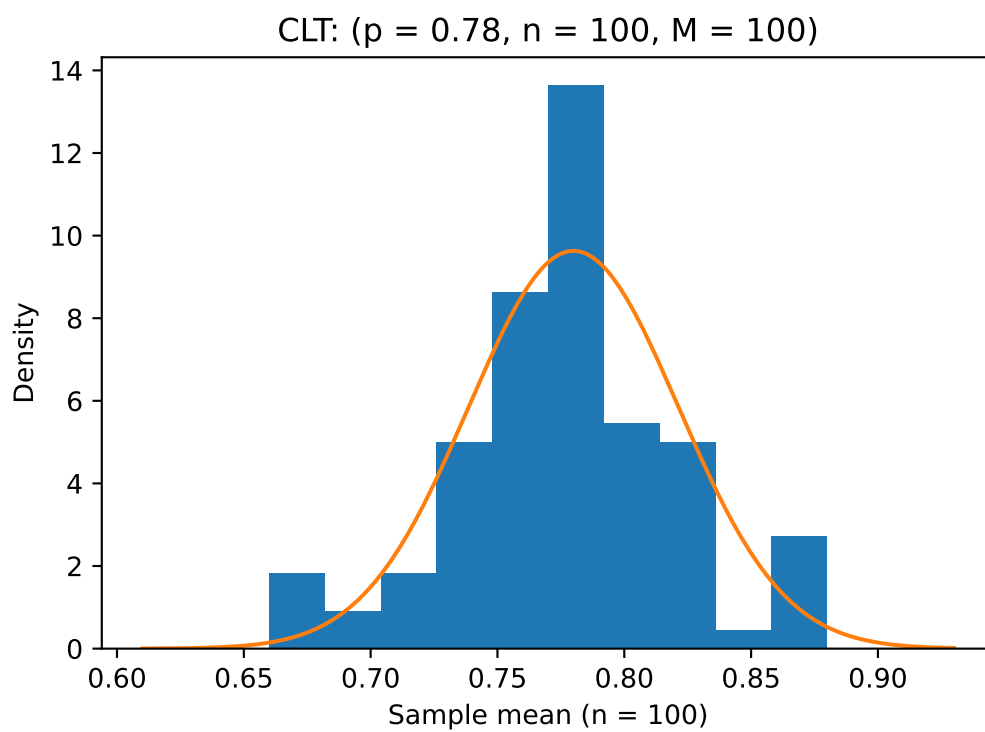
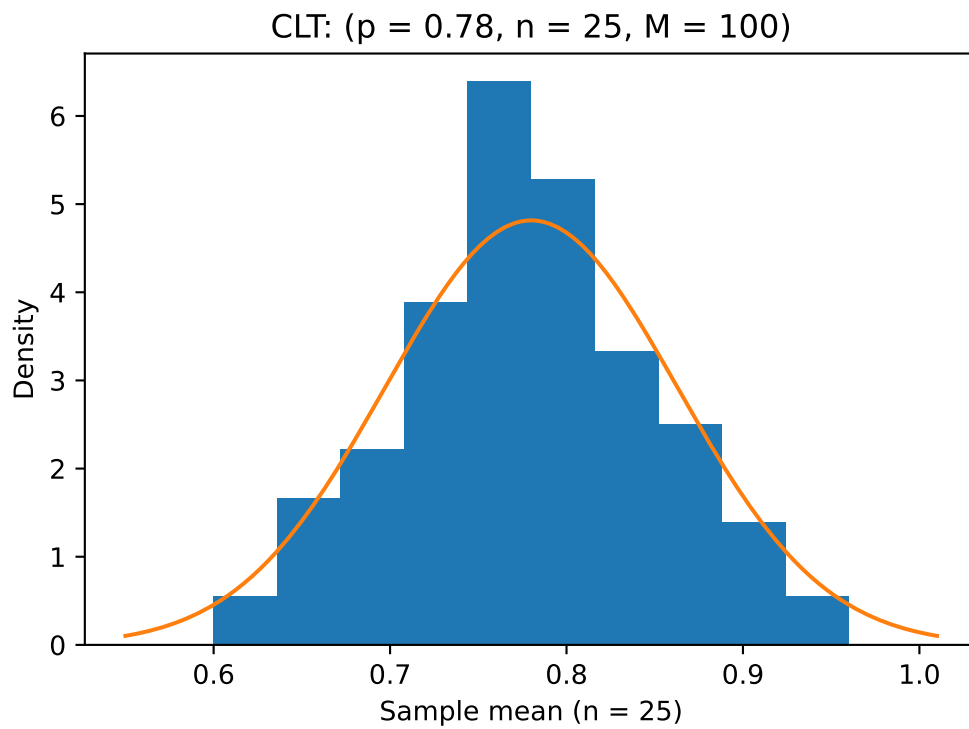
p = 0.78
M = 100
rng = np.random.default_rng(7)

def simulate_means(n):
    x = rng.binomial(1, p, size=(M, n)).mean(1)
    return x

def normal_pdf(x, mu, sd):
    return (1 / (sd * sqrt(2 * pi))) * np.exp(-(x - mu)**2 / (2 * sd**2))

for n in (25, 100):
    means = simulate_means(n)
    mu = p
    sd = sqrt(p * (1 - p) / n)

    plt.figure(figsize = (6, 4))
    # histogram of sample means (density so total area ~1)
    plt.hist(means, bins = 10, density = True)
    # theoretical normal curve
    xs = np.linspace(min(means) - 0.05, max(means) + 0.05, 200)
    plt.plot(xs, normal_pdf(xs, mu, sd))
    plt.xlabel(f"Sample mean (n = {n})")
    plt.ylabel("Density")
    plt.title(f"CLT: (p = {p}, n = {n}, M = {M})")
```



□

3c

Repeat (a) and (b) for the random draw from the standard normal distribution.

Solution Attempt:

```
M = 100
rng = np.random.default_rng(7)

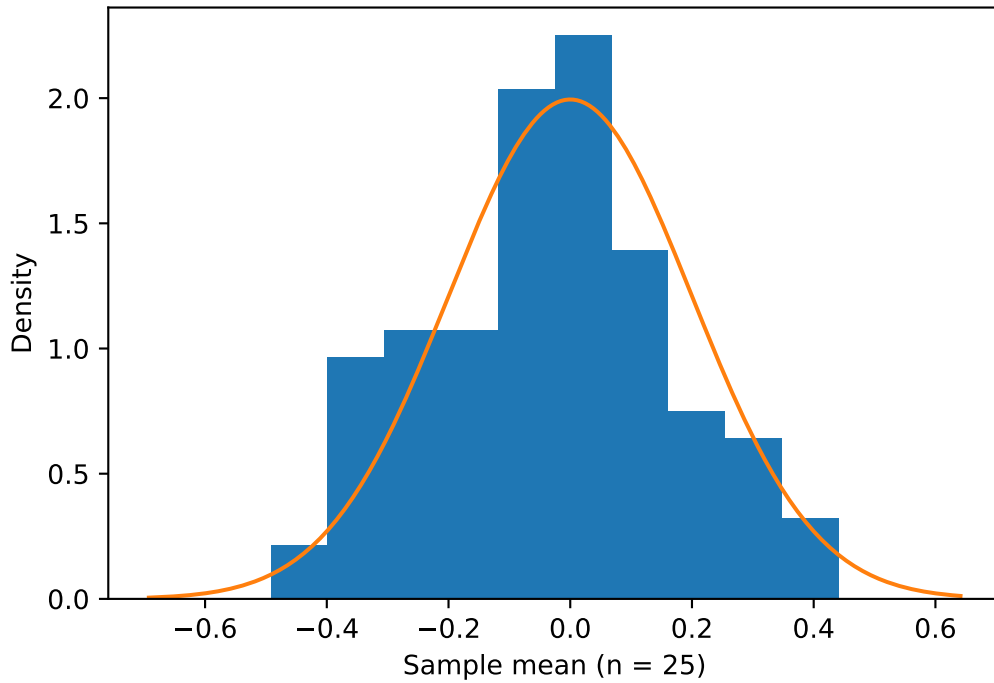
def simulate_means_normal(n):
    # M x n draws from N(0,1); take sample means
    return rng.standard_normal(size=(M, n)).mean(axis=1)

def normal_pdf(x, mu, sd):
    return (1 / (sd * sqrt(2 * pi))) * np.exp(-(x - mu)**2 / (2 * sd**2))

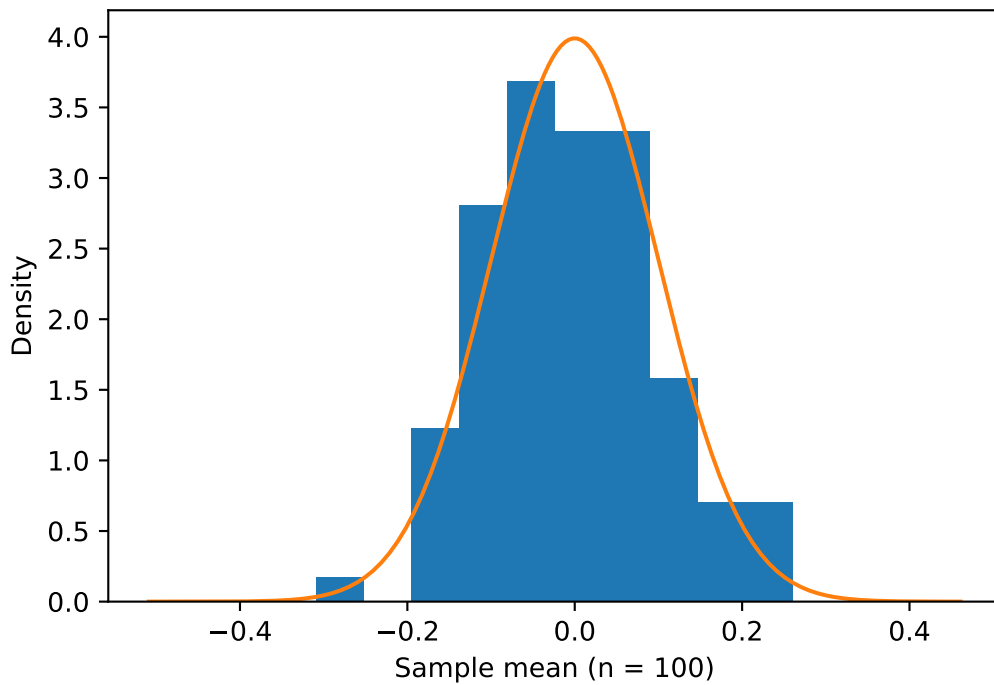
for n in (25, 100):
    means = simulate_means_normal(n)
    mu, sd = 0.0, 1.0 / sqrt(n)          # exact: mean 0, Var = 1/n

    plt.figure(figsize = (6, 4))
    plt.hist(means, bins = 10, density = True)
    xs = np.linspace(means.min() - 0.2, means.max() + 0.2, 200)
    plt.plot(xs, normal_pdf(xs, mu, sd))
    plt.xlabel(f"Sample mean (n = {n})")
    plt.ylabel("Density")
    plt.title(f"Standard Normal source: LLN/CLT (n = {n}, M = {M})")
    plt.show()
```


Standard Normal source: LLN/CLT ($n = 25$, $M = 100$)



Standard Normal source: LLN/CLT ($n = 100$, $M = 100$)



□

3d

Compare your answers for the Bernoulli and the standard normal distribution for the LLN and the CLT.

Solution Attempt:

LLN (convergence of $\bar{X}$ to the true mean)

- $Ber(p = 0.78)$: $\bar{X} \rightarrow 0.78$. Spread is approx. $\sqrt{p(1-p)/n} = \sqrt{0.1716/n}$.
- Standard Normal: $\bar{X} \rightarrow 0$. Spread is approx. $\sqrt{1/n}$.

Both concentrate at their means and shrink like $1/\sqrt{n}$. Normal shrinks from a larger variance (1 vs. 0.1716), so its histogram are wider for the same n .

CLT (shape of the sampling distribution):

- $Ber(p = 0.78)$: Approx. normal for moderate n ; with $p = 0.78$ the $n = 25$ histogram can show mild skew effects (bounded in $[0, 1]$), but by $n = 100$ it's close to normal.
- Standard Normal: $\bar{X}$ is exactly $\mathcal{N}(0, 1/n)$ for any n so the curve matches quickly.

□

Problem 4

4a

$\bar{Y}$ is an unbiased estimator of μ_Y . Is $\bar{Y}^2$ an unbiased estimator of μ_Y^2 ?

Solution Attempt:

4a

$\bar{Y}$ is an unbiased estimator of μ_Y . Is $\bar{Y}^2$ an unbiased estimator of μ_Y^2 ?